Verified Carbon Standard

A VERRA STANDARD

RED SEA POWER WIND FARM

Document prepared by ALLCOT AG

On behalf of Red Sea Power Limited SAS and SCA

Project title	Red Sea Power Wind Farm
Project ID	3163
Crediting period	01-01-2024 to 31-12-2033
Original date of issue	14/10/2024
Most recent date of issue	14/10/2024
Version	2.2
VCS Standard Version	4.7
Prepared by	ALLCOT AG

1 PROJECT DETAILS

1.1 Summary Description of the Project

The project consists of the construction and operation of a greenfield 58.9MW wind power plant near the Lake of Ghoubet, Djibouti. It involves the setting up of wind turbines which will produce electricity for export to the national grid through EDD (Electricity of Djibouti).

The wind power plant, Djibouti's first utility-scale wind park, is built and owned by Red Sea Power Limited SAS, a simplified limited liability company incorporated under the laws of the Republic of Djibouti, registered within the Commercial Registry of Djibouti Number 15366/B/SAS, a Djiboutian registered company owned by the Africa Finance Corporation (51%), the Great Horn Investment Holdings (10%)[1][1], FMO Entrepreneurial Development Bank (19.5%) and Climate Investor One (CIO) (19.5%).

Djibouti imports over 70% of its energy from Ethiopia, dominated by hydropower plants, while the local energy generation solely comes from diesel-fueled power capacities. The project will thus substitute grid electricity by renewable energy and cut down GHG emissions from baseline fossil fuel intensive grid mix. Apart from emission reductions, the benefits of the project comprise, among others, improvement of energy self-sufficiency of the country, as well as creation of local employment.

It responds to the objectives of Djibouti as stated in its INDCs committing the country to increase the share of renewables in electricity production to 100%, as echoed in the National Energy Policy of 2013 in alignment with Vision 2035.

The project's 58.9MW will generate approximately 242 GWh MWh per year and 1,017,574 tCO_2e of emission reductions over the ten-year crediting period i.e. 101,757 tC02e of emission reductions per year.

Implementation timeline:

February 2018	Feasibility studies
August 2018	ESIA
May 19th 2019	Power Purchase Agreement with Electricity de Djibouti
January 2020	Start of construction works
December 2023	Commissioning
January 2024	Commercial operation start date

Prior to the implementation of the project, the existing scenario consisted in power generation and imports from Ethiopia existing grid mix, mainly composed of hydroelectric sources (78% of the total installed capacity and power generation[2]).

1.2 Audit History

[1] Post Construction AFC will sell 10% of its share in the project to GHIH

[2] https://www.edd.dj/edd.html

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	Not applicable	VCS	Validation/verification body name	One year

1.3 Sectoral Scope and Project Type

Sectoral scope[3]	Scope 1 – Energy Industries (renewable - / non-renewable sources)
Project activity type	Renewable energy type and is not a grouped project.

1.4 Project Eligibility

1.4.1 General eligibility

The project is eligible under the scope of the VCS program version 4.7 because:

- It results in CO_2 emission reduction, one of the six Kyoto protocol greenhouse gases
- It is supported by methodologies approved under a VCS approved GHG program
- The project activity, grid-connected electricity generation using (large scale) wind power in a Least Developed Country, is not excluded as per para 2.1.3 of the VCS Standard v4.7.

Project capacity is 58.9 MW, it is a large-scale project not fragmented, therefore project is eligible under the selected methodology (ACM0002 v 22.0).

Non-AFOLU projects shall complete validation within two years of the project start date. Start date of the project is 01/01/2024.

1.4.2 AFOLU project eligibility

Not applicable since project is not an AFOLU project.

1.4.3 Transfer project eligibility

Project has not been transferred.

1.5 Project Design

[3] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes*

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

Project is not a grouped project.

1.6 Project Proponent

Provide contact information for the project proponent(s). Copy and paste the table as needed.

Organization name	Red Sea Power Limited SAS
Contact person	Sameh Fathi Elalfi Ahmed
Title	Chief Executive Officer
Address	Moulk Centre Avenue Gandhi a Gabode 3 Republic of Djibouti
Telephone	
Email	sameh.alfy@redseapower.dj

1.7 Other Entities Involved in the Project

Organization name	Sovereign Carbon Agency SAS
Role in the project	Authorized Representative
Contact person	Jean Yves KOTTO
Title	Director
Address	Zone Industrielle Sud – lot 188, P.O. Box 4476, Djibouti
Telephone	+253 77503383
Email	jykotto@sovereign-carbon-agency.com

1.8 Ownership

The legal right to control and operate the project activity belongs to Red Sea Power Limited SAS, as per PPA dated 19th May 2019.

Sovereign Carbon Agency has been appointed Authorized Representative Agent by an agreement on 23rd May 2024, with Red Sea Power Limited SAS granting the Authorized Representative all rights, responsibilities to act as if it were the Project Proponent and/or Registry User.

1.9 Project Start Date

Project start date	01-January-2024
Justification	Commercial operation start date

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	01-January-2024 to 31-December-2033

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
2024	242,000
2025	242,000
2026	242,000
2027	242,000
2028	242,000
2029	242,000
2030	242,000
2031	242,000

2032	242,000
2033	242,000
Total estimated ERRs during the first or fixed crediting period	2,420,000
Total number of years	10
Average annual ERRs	242,000

1.12 Description of the Project Activity

The proposed project consists of setting up 17 G132-3.465MW T84 (capacity of 3.465 MW each). Generated electricity is fed via aerial collector lines (i.e. cables) to a substation on the Project site. An overhead transmission line connects the Project substation (located adjacent to the RN9 at approximately lat: 11.531628 long: 42.496840) to the 230 kV circuits located in the proposed EDD Ghoubet transformer/ substation, adjacent to the RN9 national road, 3.5 km from the Project site.

This clean energy supply thus implies a substantial power capacity increase for the country, whose totally dependent of the Ethiopian grid.

The main elements of the wind power plant are 17 G132-3.465MW T84 wind turbines of 3.465 MW each; whose specifications are detailed below:

Make	Siemens Gamesa
Model	G132
Power	3.465 MW
Cut-in wind speed [m/s]	3m/s
Cut-out wind speed [m/s]	25m/s
Rated wind speed [m/s]	14m/s
Rotor diameter [m]	132 m
Converter voltage [V]	• 690 V at the output of the wind turbine generator/converter • 20 kV at the output of the wind turbine transformer
Hub height [m]	84 m
Lifetime	20 years

Geo location	COORDINATES OF WIND TURBINES **COORDINATE SYSTEM** UTM WGS84 ZONE 38 N

WTG	COORDINATE X	COORDINATE Y
WT 01	228988	1275041
WT 02	228839	1274670
WT 03	228686	1274301
WT 04	228419	1275355
WT 05	228074	1274792
WT 06	228169	1275919
WT 07	227810	1275366
WT 08	227548	1275947
WT 09	227178	1275554
WT 10	227104	1276351
WT 11	226876	1275989
WT 12	226329	1276881
WT 13	226453	1276425
WT 14	226079	1275968
WT 15	225738	1276849
WT 16	225693	1276439
WT 17	225571	1277291

Associated with 2*60MVA transformers before the metering equipment composed of 2 class 0.2S bi-directional meters ITRON SL7000-761W071 (main and check meters) for import/export located at Ghoubet 230kV sub-station Delivery Point for EDD, as well as 2 class 0.2S bi-directional meter within the wind farm's premises for control purposes.

In accordance with the provisions of the Law No. 88/AN/15/7th Regulating the Activities of Independent Electricity Producers dated 01st of July 2015, Red Sea Power Ltd has been granted a Generation License by the Ministère de l'Energie et des Resources Naturales (MERN).

1.13 Project Location

The project is located between Lake Assal and Lake Ghoubet (1 km West of the lake) in the Arta region of the Republic of Djibouti.

Its central geo coordinates are approximately 11.5300000, 42.5000000 (38N coordinate system) or 11°31'48" N – 42°30'0"

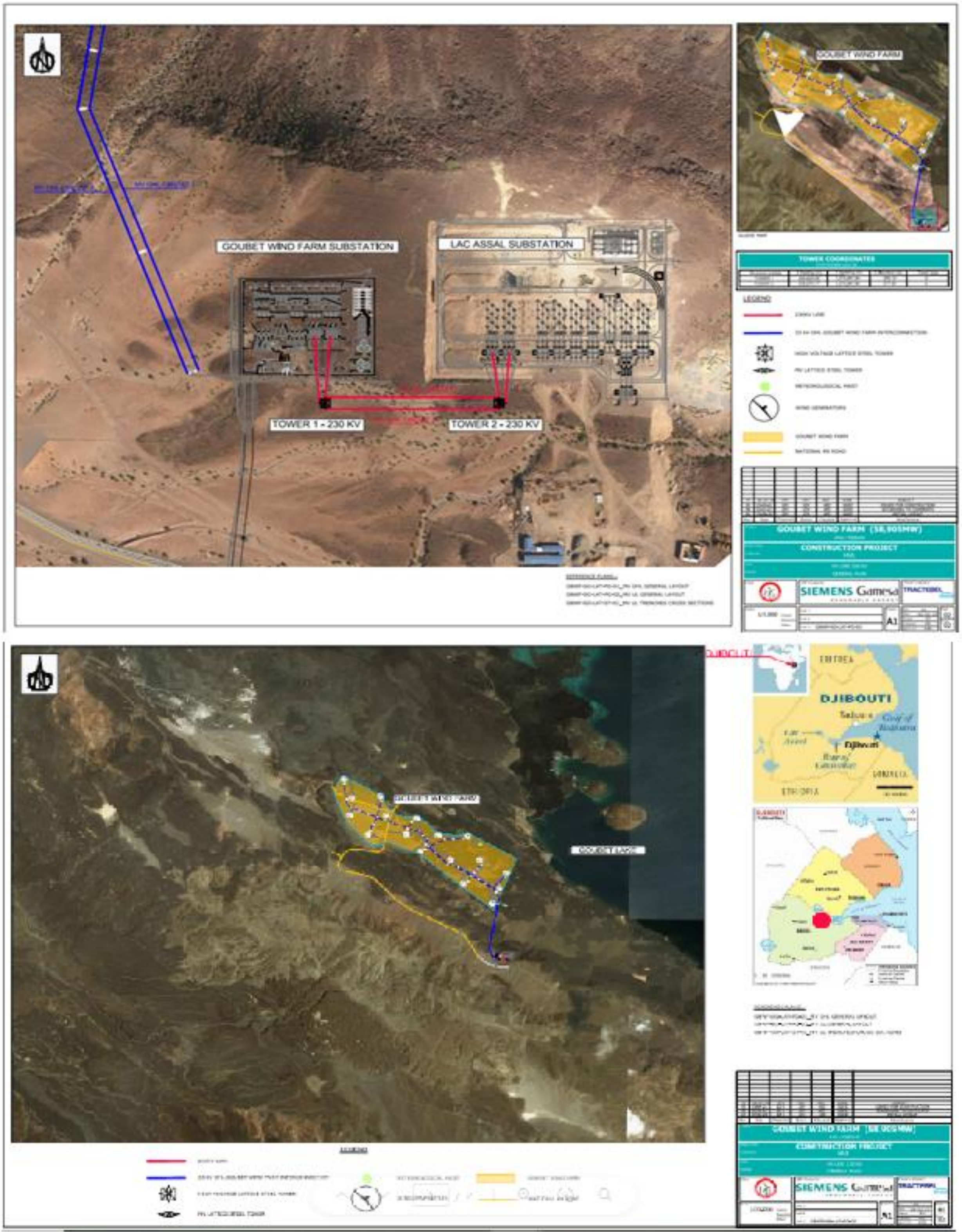


Figure 1. Site location of the project.

1.14 Conditions Prior to Project Initiation

Djibouti's electricity is primarily generated by conventional thermal power stations in the country. However, with no proven reserves of oil or natural gas and no known coal reserves, electricity being produced is dependent on imported oil and therefore susceptible to price fluctuations. Djibouti also imports hydroelectric energy from Ethiopia via a high-voltage interconnection that has been in operation since 2012. The interconnection provides a cost-effective supply of electricity throughout the year. However, power imports are not guaranteed because Ethiopia regularly curtails supply, particularly during its dry season (September to February). During the rainy season (June to August), failures of the Ethiopian transport network often led to unplanned interruptions. With national energy demand increasing between 3-5% annually, the country needs to develop new, more sustainable methods of energy production.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

To date, there are no regulations and policies preventing the implementation of the project activity, which complies with all and any relevant local, regional and national laws, statutes and regulatory frameworks:

Décret n°2009-0218/PR/MERN Establishing the national Energy. This decree establishes the National Energy Commission, whose mission is to ensure the coordination of energy projects, and more generally to undertake studies of all the measures contributing to a better coordination of the country's energy development.

This Commission is responsible for intervening in the strategic areas of energy development in the Republic of Djibouti including studies, prospecting, research, exploration, exploitation and commercial.

Such a plant, like other renewable energy power plants planned, will partly solve the problem of energy and achieve the objectives of a 100% electricity production from renewable energies goal by 2035.

In Djibouti, the procedure for evaluating impact assessments is governed by Djiboutian Government Law n°51/AN/09/6ème L bearing the code of the environment establishing the basic rules and fundamental principles of national policy in the field of environmental protection and management.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

If yes, provide the registration number and the date of project inactivity under the other GHG program.

Is the project active under the other program?

☐ Yes ☒ No

Project proponents, or their authorized representative, must attest that the project is no longer active in the other GHG program in the Registration Representation.

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

If yes, provide the program name(s), reason(s) and date for the rejection, justification of eligibility under the VCS Program, and any other relevant information.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☐ Yes ☒ No

If yes to all:

Provide evidence of the public statement. Evidence must be provided in this section or in an appendix.

1.18 Sustainable Development Contributions

Besides emission reductions impacts and fuel cost savings, economic and social benefits of this wind power project to the local population include around 403 jobs created during construction, of which 8 permanent positions[4] (local technicians, finance and administration) for the operation and maintenance of the plant. (SDG 8).

Sustainable development, supported by the social, economic and environmental pillars, is the foundation of the National Energy Policy. In practice, (renewable) energy generation contributes to the improvement of the lives and livelihoods of Djiboutian’s people, besides the nationally stated goal of a 100% electricity production from renewable energies by 2035. (SDG 7)

Energy also supports the industrialisation of Djibouti and promotes local content and local participation to satisfy the Policy’s economic requirements.

Within Red Sea, a gender action plan has been implemented, using a conformance checklist for Women’s Empowerment Principle. The actions identified as required to address the specific challenges faced by women in the Project Area of Interest and for mainstreaming gender in all aspects of the Project’s activities are presented as a Gender Action Plan (GAP). These plans are contributing to SDG 5. Achieve gender equality and empower all women and girls, especially in targets 5.1 End all forms of discrimination

[4] As per Allianz EHP Membership Census Form 3

against all women and girls everywhere; and 5.5 Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in political, economic and public life.

Fresh water resources in the Lac Assal District of Djibouti are scarce. The current supply of potable water is trucked to the Lac Assal and Moumina villages regularly by the government and stored in purpose-built structures. Red Sea Power invests in a project that will provide a reliable potable water supply to the Lac Assal and Moumina villages through water abstraction from Ghoubet Lac, desalination plant, and distribution via a pipeline. This desalination plant is contributing to SDG 6. Clean Water and Sanitation, specially target 6.1. By 2030, achieve universal and equitable access to safe and affordable drinking water for all.

Renewable energy generation implies emission reductions within the country (SDG 13), since the energy produced by the project would be otherwise produced with non-renewable sources.

Finally, project also contributes to SDG 15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss. This is done through a Biodiversity Management Plant (BMP), that covers only the work activities that are required to be carried out by the Red Sea Wind Farm during operations after full commissioning on the 1st of January 2024. The objective of this BMP is to reduce the impact of the Project's operational activities on biodiversity and the area surrounding the Project through:

- Identifying the key biodiversity issues during the project's operational phase;
- Developing strategies to manage impacts on biodiversity and implementing those strategies;
- Assigning responsibilities for impact monitoring and management;
- Providing sufficient information to assist with auditing the implementation of the BMP; and
- Establishing and implementing a biodiversity monitoring program and management measures.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

The project does not generate any leakage.

1.19.2 Commercially Sensitive Information

There is no information deemed as commercially sensitive for this project activity.

1.19.3 Further Information

The project participants obtained all necessary clearances and licenses; hence no legislative, economic, sectoral, social, environmental, geographic, site-specific risks are anticipated which may have impact on the eligibility of the project activity and the net GHG emission reductions.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	Project Stakeholders can be divided into Project affected parties (PAPs) and Other Interested Parties (OIPs). • PAPs include people, groups and other entities within the Project Area of Interest (AoI) that are directly influenced (actually or potentially, positively or adversely) by the Project and/or have been identified as most susceptible to change associated with the Project. They should be closely engaged in identifying impacts and their significance, as well as in decision-making on mitigation and management measures. • OIPs include individuals/groups/entities that may not experience direct impacts from the Project but who consider or perceive their interests as being affected by the Project and/or who could affect the Project and the process of its implementation in some way. This SEP outlines the needs and expectations for engagement of these stakeholders with appropriate methods developed to meet these needs and interests and ensures that their views and concerns are addressed in a suitable manner. **RSP (REd Sea Power) has adopted an iterative approach to stakeholder identification and engagement that follows the overall objectives and approach presented in Figure 7.1** of Red Sea Power Project STAKEHOLDER ENGAGEMENT PLAN. Steps are: Identify, understand, inform, relationship, trust, engage, manage and compliance.

	Based on the above approach, the Project has identified a diverse range of stakeholders during the ESIA process. These include, but are not limited to, representatives of National Government Stakeholders, Local / District Authority Stakeholders, Community Stakeholders / Traditional leaders, NGOs, Associations and Other Organizations and Individuals
Legal or customary tenure/access rights	There are some stakeholders who use land with no legal binding: • non-registered land users or those who use land on the basis of traditional non-legal claims • traditional nomadic families migrating to remote areas needing tailored channels of communication; • traditional leaders of the communities using the lands and natural resources in the Project locality [5]
Stakeholder diversity and changes over time	N/A.
Expected changes in well-being	A desalination plant will be installed to provide water to the citizens of the area of interest.
Location of stakeholders	Lac Assal and Cire Mounima and other places close to the Area of Interest.
Location of resources	The area of interest.

2.1.2 Stakeholder Consultation and Ongoing Communication

The following meetings were held with stakeholder, during the ESIA process from February to December 2018.

[5] Red Sea Power Project STAKEHOLDER ENGAGEMENT PLAN

Stakeholder Engagement Meetings Held

Date	*Stakeholder*	*Participants*		
		Male	*Female*	*Total*
Government meetings				
12 Dec 2018 17-19 Feb 2018 3 May 2018	Electricté de Djibouti (EDD)	1		1
11 Feb 2018	Ministry of Agriculture	1		1
26 Feb 2018	Ministry of Habitat, Urban Planning, Environment and Town Planning (MHUE)	1		1
2 May 2018	Environment and Sustainable Development Directorate (part of MHUE)	2		2
	Total	*5*		*5*
Local level meetings				
8 Feb 2018	Arta Prefecture, meeting with Prefect	1		1
10 Feb 2018	Cité Moumina, Focus Group discussion with customary authorities	2		2
10 Feb 2018	Karta authority	1		1
11 Feb 2018	Tadjourah Prefecture, meeting with Prefect	1		1
14 Feb 2018	Public consultation in Lac Assal village. Attendees included: Sub-prefect and Village Chief of Lac Assal, members of local associations (including the Women's and Youth associations)	11	2	13
17 Feb 2018	Meeting with Okal, customary authority	1		1
17 Feb 2018	Public consultation in Cité Moumina community. Attendees included customary authorities and village elders.	6		6
19 Feb 2018	Focus Group discussion, Cité Moumina / Lac Assal Women's Assocation		2	2
18-19 Feb 2018	Individual interviews with Key informants (such as livestock breeders, fishermen, manager of the tourist resort) during the social baseline engagement.	7		7
15-19 Feb 2018	Household surveys were completed with 40 households in the Project area during the social baseline engagement.			40
3 May 2018	Public consultation in Lac Assal's Community Building. Attendees included: Sub-prefect and Chief of Lac Assal, Lac Assal Women's Assocation, Okal General, Imam of Cité Moumina Mosque, Makaban (customary authorities representing the Debné tribes) and community members.	14	2	
	Total	*30*	*6*	*74*
	Grand Total	***35***	***6***	***79***

Date of stakeholder consultation	Several dates, described in the table above.
Stakeholder engagement process	RSP is committed to engaging with its stakeholders in a culturally appropriate manner and in a language that they understand. A

	variety of communication methods have therefore been selected for Project engagement activities, based on the level and objective of the engagement, as well as the target group, as shown in Table 7.1. of the Red Sea Power Project STAKEHOLDER ENGAGEMENT PLAN (SEP). Further modes of engagement may be introduced if identified as required during subsequent stages of the Project. Table 7.2 of the same document shows Methods, Tools and Techniques for Stakeholder Engagement.
Consultation outcome	During the public consultation meetings discussions with stakeholders included: • presentations of the Project; and • discussions on the potential impacts of the Project, on the proposed solutions to minimise negative impacts and maximise positive benefits, and on the overall expectations regarding the project.
Ongoing communication	An effective grievance mechanism allows stakeholders to lodge complaints and/or concerns at no cost, without retribution and with the assurance of a timely response. The process will include the following steps: • identification; • review and record the grievance; • acknowledgement; • develop a response; • communicate response and establish agreement; and • close-out process. A detailed description of the grievance mechanism required for the Project is provided in the SEP. Stakeholders can submit grievances or comments regarding the Project to the Community Liaison Officer or through local community leaders. Prior to construction, an office was established at the site to improve accessibility of the grievance process.

Stakeholder input	Describe how due account was taken of all input received during the consultation. Include details on any updates to the project design or justify why updates were not necessary or appropriate.

2.1.3 Free Prior and Informed Consent

Use the table below to describe the outcome of the FPIC process as part of the stakeholder consultation process.

Obtaining consent	**An effective SEP was carried on, Effective Stakeholder Engagement requires an integrated approach between the Company, Contractors (and any subcontractors) and regulatory authorities, amongst others. This requires robust processes to be in place regarding information dissemination, training, designation of responsibility, management actions, monitoring, control, and corrective actions. Ownership of and ultimate responsibility for this document lies with RSP, but much of its daily implementation will be managed by the EPC contractor during construction and the O&M contractor during operations.**	An effective SEP was carried on, Effective Stakeholder Engagement requires an integrated approach between the Company, Contractors (and any subcontractors) and regulatory authorities, amongst others. This requires robust processes to be in place regarding information dissemination, training, designation of responsibility, management actions, monitoring, control, and corrective actions. Ownership of and ultimate responsibility for this document lies with RSP, but much of its daily implementation will be managed by the EPC contractor during construction and the O&M contractor during operations.
Outcome of FPIC	RSP made consultation with parties throughout different meetings, as it is explained above. Stakeholders expressed their willingness to collaborate for the project success.	

2.1.4 Grievance Redress Procedure

Use the table below to describe the grievance redress procedures developed to resolve any conflicts which may arise between the project proponent and stakeholders.

Development process	An effective grievance mechanism allows stakeholders to lodge complaints and/or concerns at no cost, without retribution and with the assurance of a timely response. The process will include the following steps: • identification;

	• review and record the grievance; • acknowledgement; • develop a response; • communicate response and establish agreement; and • close-out process. A detailed description of the grievance mechanism required for the Project is provided in the SEP.
Grievance redress procedure	Stakeholders can submit grievances or comments regarding the Project to the Community Liaison Officer or through local community leaders. Prior to construction, an office was established at the site to improve accessibility of the grievance process. The Grievance Mechanism serves to: • Minimize any adverse impacts of the Project on external stakeholders, via the prompt and mutually acceptable resolution of grievances; • Identify emerging adverse trends in terms of incidents/impacts at an early stage so that measures to prevent/avoid their occurrence can be implemented quickly and proactively; • Demonstrate the Project's respect for the interests of external stakeholders. The key principles of the grievance management process are that: • Any person, group or organization can submit a grievance at any time, without fear of retribution and at no financial cost. • All grievances will be taken seriously and will be treated in a fair and respectful manner. The Company will respond to the complainant to confirm receipt of the grievance within seven working days. • The process by which grievances will be received, investigated and resolved will be consistent and transparent. Representatives of contractors may be involved in the investigation where applicable.

	• Information relating to grievance investigations and eventual decisions will be documented. • Complainants will have recourse to an internal Project appeal mechanism if the complainant rejects the (first) decision. • Personal information about the affected stakeholders will be treated as confidential. • The mechanism will not impede access to judicial or administrative remedies. The Grievance Mechanism consists of 6 main stages namely: • Stage 1: Grievance communication and logging (registration) • Stage 2: Acceptance of grievance for investigation • Stage 3: Notification • Stage 4. Investigation • Stage 5: Resolution Stage 6: Monitoring and evaluation

2.1.5 Public Comments

The following table provides a summary of issues raised and how they have been addressed in the ESIA :

Subject	Issue	Project Response	ESIA Reference
Land Use and Land Take	• The Project footprint does not constitute a major issue for the community. The affected land is unsuitable for any productive activity, and the Project is thus not expected to hinder any agricultural or pastoral projects of the residents. • Some concerns were raised about the project actually materialising: reportedly projects have been planned in the past with no follow through.	• Access will be temporarily restricted in small areas of the site during construction. The Project will engage with communities beforehand to minimise the impact and agree on alternative accessible routes. • The Project will employ a Community Liaison Officer who will communicate all Project developments to the communities so that they are informed of Project development progress.	*Section 7.12 (Land Use and Livelihoods)*
Employment and the Economy	• Community members have high expectations in terms of access to jobs, both during the construction phase of the project and as a result of the electrification of the area. • Some members expressed concerns that outsiders will take the jobs, due to the lack of schooling and professional training of the local population – and this will reportedly be unacceptable for them.	• The Project will implement a Human Resources Policy and Management Plan setting out recruitment procedures, training requirements and targets around the hiring of local workers. • Local procurement will be prioritised where possible. • The Project will provide information on the specific jobs available and specific skills required, at least two months prior to the hiring period. This is to ensure opportunities are known in advance for unskilled and semi-skilled positions, which are more likely to be filled by local residents.	*Section 7.11 (Employment, Procurement and Economy)*
Demography and in-migration	• In Lac Assal, potential demographic increase as a result of the project is seen as an opportunity. Some concerns were raised related to unplanned or chaotic urbanisation, and waste management. • In Cité Moumina, the community members maintained that the presence of workers or jobseekers from other areas will be a concern for them, and thus the village elders should be updated continuously to manage any potential influx of people.	• The Project will aim to detract work-seeking in-migration to the Project area through the clear communication of Project labour needs and clear Project policies and procedures for recruitment. • The Project will implement an Influx Monitoring Programme and an Influx Management Plan (if influx is identified as a growing concern during monitoring programme) • The Project will implement a Grievance Mechanims that is easily accessible to local communities	*Section 7.13.7 (Community Health, Safety and Security)*
Environment	• The community members were given the opportunity to ask questions regarding windfarms - noise pollution was the theme that raised most concerns.	• Noise emissions from construction will be temporary and limited to the duration of the equipment operations at the work sites. • It is planned that turbines will be sited at least 500 m from any dwellings to mitigate noise impacts. • The Project will commit to the IFC EHS General Guidelines on noise impacts: 45 dB(A) limit during operation at residential receptors.	*Section 7.10 (Noise)*
Health and wellbeing	• No particular concerns related to health and safety during the construction phase, especially if local people were to be in charge of site safety. • Questions related to the safety of the area once the turbines have been installed (lightening and fears that the rotor blades may detach and fall).	• The Project will engage with the local community through the Community Liaison Officer about the risks and safety features of the wind turbines; turbine failure resulting in 'turbine throw' is considered unlikely. • Set-back distances (≥500 m) between turbines and residential areas will be incorporated into the design. • The Project will engage with the local community on what to do in the event of an emergency scenario such as turbine failure.	*Section 7.13 (Community Health, Safety & Security)* *Section 8 (Unplanned Events)*
Cultural heritage	• There were concerns that the cemeteries located just outside the project perimeter will be damaged by the Project.	• The cemeteries are located outwith the Project site therefore direct impacts to them will be avoided. A suitable buffer has also been included in the turbine layout design.	*Section 6.10 (Cultural Heritage)*
Engagement during the Project	• Recommended means to set up a good communication and information sharing system between the project and the local communities.	• During the public consultation meetings, the EDD representative noted that there are plans for weekly meetings with the community. This plan was welcomed by the community. • The Project will employ a Community Liaison Officer who will communicate all Project developments to the communities. • A community grievance mechanism has been communicated and participants welcomed this.	*Annex G Stakeholder Engagement Plan*

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

SEP will be managed and controlled in line with the RSP Document Control Processes, which is available in the HSSE-Management Systems Framework Document.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	Activities will have a direct impact on local community safety and security.	The Security Management Plan will remain in place for operation as well as the Traffic Management Plan. A Community Liaison Officer (CLO) will also be present to engage with local communities. A community development plan will also be in place to support local community development priorities. With these measures in place, impact significance is predicted to be Not Significant during operation.
Risks to stakeholder participation	Risks to nearby communities	During ESIA Process, key stakeholders, including interested and affected parties, were identified and provided with an opportunity to raise any comments, concerns /and/or queries that they may have on the proposed project. The Stakeholder Engagement Plan recognizes that stakeholder engagement is fundamental to good business practice; and, at a community level, is central to both managing risks and impacts to the nearby communities and enhancing potential benefits.
Working conditions	Djibouti has ratified all eight core International Labour Organization (ILO) conventions, and a legal framework is in place to protect	A contractual management system will be established including policies and codes of practices, which dictate rules of

	salaried workers, however local workers in Djibouti are vulnerable to poor working conditions. Migrant populations are also particularly vulnerable to exploitation, forced labor and trafficking, which must be considered in the procurement supply chain and subcontracting arrangements. Baseline findings suggest that adequate working conditions in terms of working hours and rest times are not always respected locally. Due to the context of labor and working conditions and human rights in Djibouti, receptors are considered to have High vulnerability to impacts that affect wellbeing and health and safety and worker rights in the absence of any active mitigation measures. Workers will also be vulnerable during retrenchment at the end of the construction period, particularly if they do not understand their rights with respect to retrenchment.	engagement for sub-contractors working on behalf of the Project. Expectations on labor and working conditions will be made clear at the bidding stage and will be clearly articulated in procurement and contract documentation.
Safety of women and girls	Unsafety conditions for women workers.	Gender Analysis and Action Plan together with Community Health and Safety Plan. Project adheres to Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) (1981). Within Red Sea, a gender action plan has been implemented, using a conformance checklist for Women's Empowerment Principle. The actions identified as required to address the specific challenges

		faced by women in the Project Area of Interest and for mainstreaming gender in all aspects of the Project's activities are presented as a Gender Action Plan (GAP).
Safety of minority and marginalized groups, including children	Potential risk of marginalization.	Company adheres to Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) (1981) Convention on the Rights of the Child (1990)
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Surface water - contamination. Pollution of waste is not properly disposed. The Project will cause localized, temporary impacts on air quality due to on-site construction activities. Under normal operations there will be no gaseous emissions from the operating areas. For more info, please consult ESIA Report, Volume II.	According to EIA: A Waste Management Plan was developed during the detailed design phase and implemented as soon as site preparation works commenced. Fugitive dust emissions arising from various activities such as excavation, of material (loading and unloading), and vehicular movement (on unpaved roads) were minimized through construction site good practice. There were gaseous and fugitive dust emissions owing to movement of maintenance vehicles, which were minimized through onsite speed limits of 10-15 kph. Vehicular emissions are controlled through proper vehicle maintenance.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified[6]	Mitigation or preventative measure(s) taken
Discrimination	No risk identified in the project.	According to EIA, The Afar people, who inhabit the region that the Project site is located within, are an ethnic majority in Djibouti (~30% of total population) and are not considered to be subject to particular discrimination. They have not been the object of conquest and domination therefore they are holders of the power on their territory. They are not enslaved, stigmatized or marginalized by another ethno-linguistic group. The Afar people are therefore not considered to be an indigenous people. Human resources policy of Red Sea, afford equal opportunity for employment to all potential and existing employees without regard to race, creed, color, national origin, religion, gender, sexual orientation or any other basis, as protected by international conventions and/or national law.
Sexual harassment	Women are faced with specific health and safety risks. Gender-based violence and harassment (GBVH) can be particularly pervasive in societies where there is pronounced gender inequality and power imbalances. No risk identified in the project.	The Project must take steps to ensure that gender-based health and safety needs and risks are considered and adequately addressed according to Gender Analysis and Action Plan.
Equal pay for equal work	No risk identified in the project.	The Labour Code (2006) protects all people from discrimination when seeking employment and

[6] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

		requires equal payment for men and women. Human resources policy of Red Sea, applies equal pay for equal work principle.
Gender equity in labor and work	No risk identified in the project.	The Project must take action to ensure that women are equally informed about and empowered to participate in and profit from the direct and indirect opportunities and benefits presented by the Project. Human resources policy of Red Sea, afford equal opportunity for employment to all potential and existing employees without regard to race, creed, color, national origin, religion, gender, sexual orientation or any other basis, as protected by international conventions and/or national law.
Forced labor	No risk identified in the project.	Human resources policy of Red Sea, states "Not employ or engage forced, bonded or child labor, and respect national law and the International Labour Organisation (ILO) requirements on minimum age for workers."
Child labor	No risk identified in the project.	Child labour (under 16 years old) and night work for youth (under 18 years old) are forbidden (Title I, Art. 5 and Title III, Chapter 1, Section 2, Art. 94, GdD 2005). Human resources policy of Red Sea, states "Not employ or engage forced, bonded or child labor, and respect national law and ILO requirements on minimum age for workers."
Human trafficking	No risk identified in the project.	Human resources policy of Red Sea, states "Not employ or engage forced, bonded or child labor, and

		respect national law and the International Labour Organisation (ILO) requirements on minimum age for workers.”

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified in the project.	The republic of Djibouti has ratified or adheres to the main Human Rights Instruments. Human resources policy of Red Sea aims to provide their employees with information regarding their rights under national labor and employment law, including their rights related to wages, benefits and to form unions.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No-risk identified	Applying World Bank Group Safeguard Policies according to EIA. The EA Policy takes into account the natural environment (air, water, and land), human health and safety, social aspects (involuntary resettlement, indigenous peoples, and physical cultural resources), as well as transboundary and global environmental aspects.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No-risk identified	According to EIA, Vol II: The Directorate is in charge of all expropriation operations within the country and land registration or regularization in the case of small scale compensation projects.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Stakeholder engagement plan includes sharing of development benefits and opportunities. Vulnerable persons may also be disadvantaged in sharing development benefits and opportunities and specific steps may be needed to access these groups and afford them the opportunity to engage in discussion about the Project and their interactions with it. It provides a framework for stakeholder engagement throughout the life of the Project (planning, construction and operation) that is effective, meaningful, consistent, comprehensive, coordinated and culturally appropriate. Since stakeholder engagement will be an ongoing process throughout the lifetime of the Project, this SEP is a “living document” and will be updated and adjusted as the Project progresses.2 It recognises that stakeholder engagement is fundamental to good business practice; and, at a community level, is central to both managing risks and impacts to the nearby communities and enhancing potential benefits
Summary of the benefit sharing plan	Local Content Plan was developed to formalize the commitments and procedures for recruitment of local candidates for employment by the Project. This plan has been prepared in alignment with the Human Resources (HR) policies of the constructor. The constructor implemented a strategy to increase local procurement, the award of contracts for the supply of goods and services to local businesses contributes to the sharing of Project benefits with local stakeholders through the stimulation of economic activities in new sectors and industries as well as the creation of indirect jobs. Local procurement is defined as the purchase of goods and services from business or individual providers that have a registered business and their business headquarters (if applicable) in Djibouti. In cases of equal qualification and value, suppliers from the local level are contracted first, followed by those from the regional level, followed by those from the national level. Red Sea Power invests in a project that provides a reliable potable water supply to the Lac Assal and Moumina villages through water abstraction from Ghoubet Lac, desalination plant, and distribution via a pipeline.
Approval and dissemination of	*Demonstrate that the benefit- sharing agreement was agreed up on by the affected stakeholder groups, and that the agreement was shared*

benefit sharing plan	*in a culturally appropriate manner. Demonstrate that the agreement is readily accessible should stakeholders wish to review the agreement.*

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	The Project involved localized vehicle movements from within Djibouti, minimizing the likelihood of non-native species being introduced via equipment, vehicles and personal belongings of the workforce. No invasive alien species were recorded during baseline surveys.	The following measures and procedures were implemented during the construction of the Project: • Check origin of equipment and materials used on site; and • Implement invasive species management procedure as part of BAP, including cleaning of equipment prior to shipping to site. The turbines and other project components will have passed through import checks prior to being cleared for movement to the site. Taking this into account, and with the mitigation measures set out above, the impact on receptors will be reduced to negligible and the effect is considered to be Not Significant.
Soil degradation and soil erosion	Groundworks and construction activity resulting in soil compaction and loss of soil stabilizing vegetation (therefore increasing surface runoff and localized erosion).	EIA states how project is avoiding soil degradation.
Water consumption and stress	Not risk identified.	According to EIA, A Water Resources Study was performed to confirm Office National de l'Eau et de l'Assainissement de Djibouti's (ONEAD) statement that the source

		of water used for the Project is sufficient to supply to water required for the Project and not impact supply to other parties.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes ☐ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat	According to EIA, the majority of plant and animal species recorded on site are all common and widespread. Two mammal species of conservation concern, the IUCN VU Dorcas gazelle and IUCN NT striped hyena, are considered to be of medium sensitivity/value. One reptile species, the Djibouti whip snake is currently considered to be endemic to Djibouti and is considered to be of medium sensitivity/value as a result of its known limited distribution and likely small population size. The sensitivity of all other terrestrial biodiversity resources (expect birds and bats) within Project Area of Interest (AoI) is assessed through the baseline studies as being low. This is because the habitats and majority of species present on site are common and widespread across Djibouti.
Areas needed for habitat connectivity	The habitats and majority of species present on site are common and widespread across Djibouti.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified, the habitats and majority of species present on site are common and widespread across Djibouti.	NA
Areas for habitat connectivity	No risk identified, the habitats and majority of	NA

	species present on site are common and widespread across Djibouti.	

2.4.2 Introduction of Species

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	N/A

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	N/A	N/A

2.4.3 Ecosystem Conversion

Not applicable for renewable projects.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	N/A

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	“Grid connected electricity generation from renewable sources”	22.0
Tool	TOOL01	Tool for the demonstration and assessment of additionality	7.0.0

Tool	TOOL 05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	3.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system	7.0

3.2 Applicability of Methodology

Table 1. Compliance of the project activity regarding applicability conditions of methodologies and tools.

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); Or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	The project activity is a greenfield wind power plant substituting electricity produced on the grid by renewable energy.
ACM0002	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that:	Project activity does not involve the implementation of BESS:

	(a) Integrate BESS with a Greenfield power plant; (b) (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic1 or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	
ACM0002	The methodology is applicable to the following technologies: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission	Project does not involve BESS neither, capacity addition nor hydro power plants.

section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity;

(c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents);

(d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies [2] may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.

(e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in

	coordination with the PSP during pumping mode.	
ACM0002	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m2 ; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m2 ; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m2 , and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m2 ; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m2 are:	Project is not hydro power type.

	a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
ACM0002	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water supplied to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	Project is not hydro power type.
ACM0002	In the case of PSP, the project participants shall demonstrate in the PDD that the	Project is not a PSP.

	project is not using water which would have been used to generate electricity in the baseline.	
ACM0002	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) (b) Biomass fired power plants/units.	Project involves a greenfield power plant that does not include fuel switching or biomass fired power units
ACM0002	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	Project activity does not involve retrofits, rehabilitations, replacements, or capacity additions.
TOOL01	The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool	The project participants do not propose any new methodology; thus the tool is applicable
TOOL01	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Tool 01 is included in the methodology ACM0002, thus its application by project participant is mandatory.

TOOL05	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	The project activity is to supply energy to the grid, though it may consume some also from the grid, thus scenario A is applicable. No captive power plant is installed on site.
TOOL05	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated:	As per ACM0002, V.20, “Accordingly, $EG_{facility,y,}$ $EG_{PJ\_Add,y,}$, $EC_{BESS,y}$, and $EC_{PSP,y}$ should be determined as per “TOOL05: Baseline,

	(a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities	project and/or leakage emissions from electricity consumption and monitoring of electricity generation." As per para. 27 of ACM0002, one possible project scenario is the installation of a Greenfield power plant delivering electricity to the grid, i.e. at least scenario I applies and the tool can be referred to.
TOOL05	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO_2 emissions.	No captive renewable power generation technology is installed.
TOOL07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	As part of ACM0002, "operating margin" (OM), "build margin" (BM) and "combined margin" (CM) need to be estimated to calculate baseline emissions of the project activity that substitutes electricity in the Djiboutian grid. Hence the tool is applicable.
TOOL07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off - grid power plants. In	The emission factor for the project electricity system is calculated for grid power plants and

	the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	off-grid power plants. Option IIb is applied, i.e. the tool is applicable.
TOOL07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Since the project electricity system is not located partially or totally in an Annex I country - it is located in the Republic of Djibouti - the tool is applicable.
TOOL07	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	There are no biofuels used in the project activity, i.e. the tool is applicable.

Other tools mentioned in the methodology are not applicable to this project activity.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO_2	Yes	Main emission source
		CH_4	No	Minor emission source
		N_2O	No	Minor emission source
Project	For dry or flash steam geothermal power plants, emissions of CH_4 and CO_2 from non-condensable gases contained in geothermal steam	CO_2	No	Not applicable (Only for geothermal)
		CH_4	No	Not applicable (Only for geothermal)
		N_2O	No	Not applicable (Only for geothermal)
	For binary geothermal power plants, fugitive emissions of CH_4 and CO_2 from non-condensable gases contained in geothermal steam	CO_2	No	Not applicable (Only for geothermal)
		CH_4	No	Not applicable (Only for geothermal)
		N_2O	No	Not applicable (Only for geothermal)
	For binary geothermal power plants, fugitive emissions of hydrocarbons such as n-butane and isopentane (working fluid) contained in the heat exchangers	Low GWP hydrocarbon/refrigerant	No	Not applicable (Only for geothermal)
	CO_2 emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants	CO_2	No	Not applicable (Only for solar thermal power plants and geothermal power plants)
		CH_4	No	Not applicable (Only for solar thermal power plants and geothermal power plants)
		N_2O	No	Not applicable (Only for solar thermal power plants and geothermal power plants)
	For hydro power plants, emissions of CH_4 from the reservoir	CO_2	No	Not applicable (Only for hydro)
		CH_4	No	Not applicable (Only for hydro)

Source		Gas	Included?	Justification/Explanation
		N_2O	No	Not applicable (Only for hydro)
	Charging of BESS using electricity from the grid or from fossil fuel electricity generators.	CO_2	No	Not applicable (Only for project with BESS)
		CH_4	No	Not applicable (Only for project with BESS)
		N_2O	No	Not applicable (Only for project with BESS)
	Utilization of electricity from grid or from fossil fuel generators by PSP for pumped mode.	CO_2	No	Not applicable (there is no PSP)
		CH_4	No	Not applicable
		N_2O	No	Not applicable
	For PSP, emissions of CH_4 from the reservoir	CO_2	No	Not applicable
		CH_4	No	Not applicable (there is no PSP)
		N_2O	No	Not applicable

According to ACM0002 methodology, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the VCS project power plant is connected to. The project boundary is therefore determined as:

- the project activity site, where the electricity is being produced,
- the grid that the power plant is connected to.

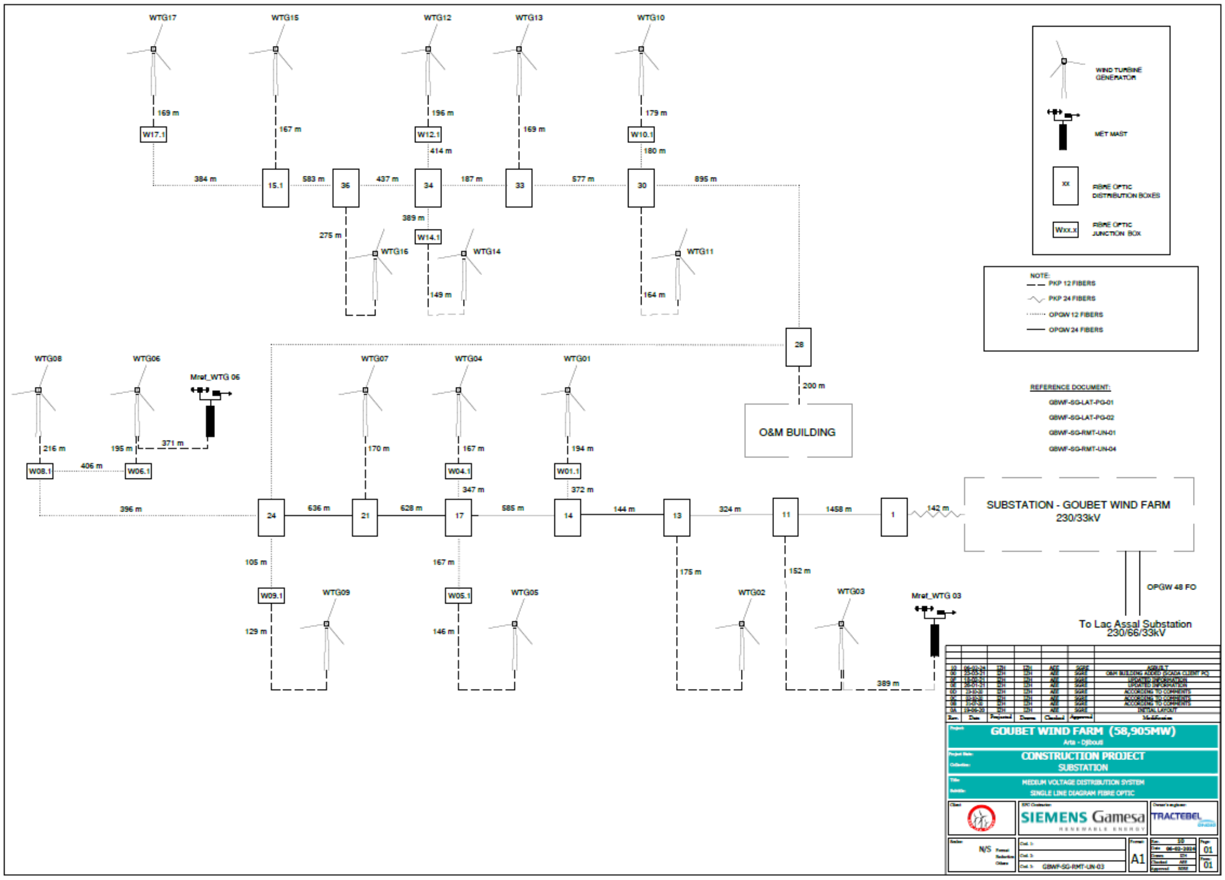


Figure 2. Project boundary.

3.4 Baseline Scenario

Identify and justify the baseline scenario, in accordance with the procedure set out in the applied methodology and any relevant tools. Where the procedure in the applied methodology involves several steps, describe how each step is applied and clearly document the outcome of each step.

Explain and justify key assumptions, rationale, and methodological choices. Provide all relevant references.

According to ACM0002 Version 22.0 and since the project activity is the installation of a new grid-connected renewable power plant (Greenfield) the baseline scenario is the following:

“The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07.”

Baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as described in section 4.1

3.5 Additionality

According to ACM0002 Version 22, the tool 01, "Tool for the demonstration and assessment of additionality" version 07.0.0 applies for additionality demonstration, as well as the Tool 23 Additionality of first of its kind project activities version 03.0.

Step 0 from the methodology procedure of the tool for the demonstration and assessment of additionality is to demonstrate whether the proposed project activity is the first of its kind.

The tool 23 Additionality of first of its kind project activities states that a proposed project activity is the first of its kind in the applicable geographic area (host country) if:

Applicable area	Entire host country Djibouti (a)
Measure	Switch of technology with change of energy source (b) power generation based on renewable energy
Output	Goods/services produced by the project activity Electricity
Different technologies	Size of installation (power capacity) Large
Crediting period	The project participants selected a crediting period for the project activity that is "a maximum of 10 years with no option of renewal" (c)
Evidence	As of project's start date (01/01/2024), no other large-scale wind power plant (>15 MW) was in commercial operation in Djibouti.

The project is thus considered additional.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☒ Yes ☐ No

If no, describe which mandated laws, statutes, or other regulatory frameworks require project activities and provide evidence of systematic non-enforcement to demonstrate regulatory surplus.

3.5.2 Additionality Methods

No additional methods.

3.6 Methodology Deviations

No methodological deviations were used.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and/or the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	Baseline emissions in year y (t CO_2/yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and supplied to the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using TOOL07 (t CO_2/MWh)

Calculation of $EG_{PJ,y}$

Since the project activity consists of the installation of new grid-connected renewable power plant at a site where no renewable power plant was operated prior to the implementation of the project activity, the project activity is the installation of a greenfield renewable energy power plant, so that:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$	Quantity of net electricity generation that is produced and supplied to the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $\mathbf{EF_{grid,CM,y}}$

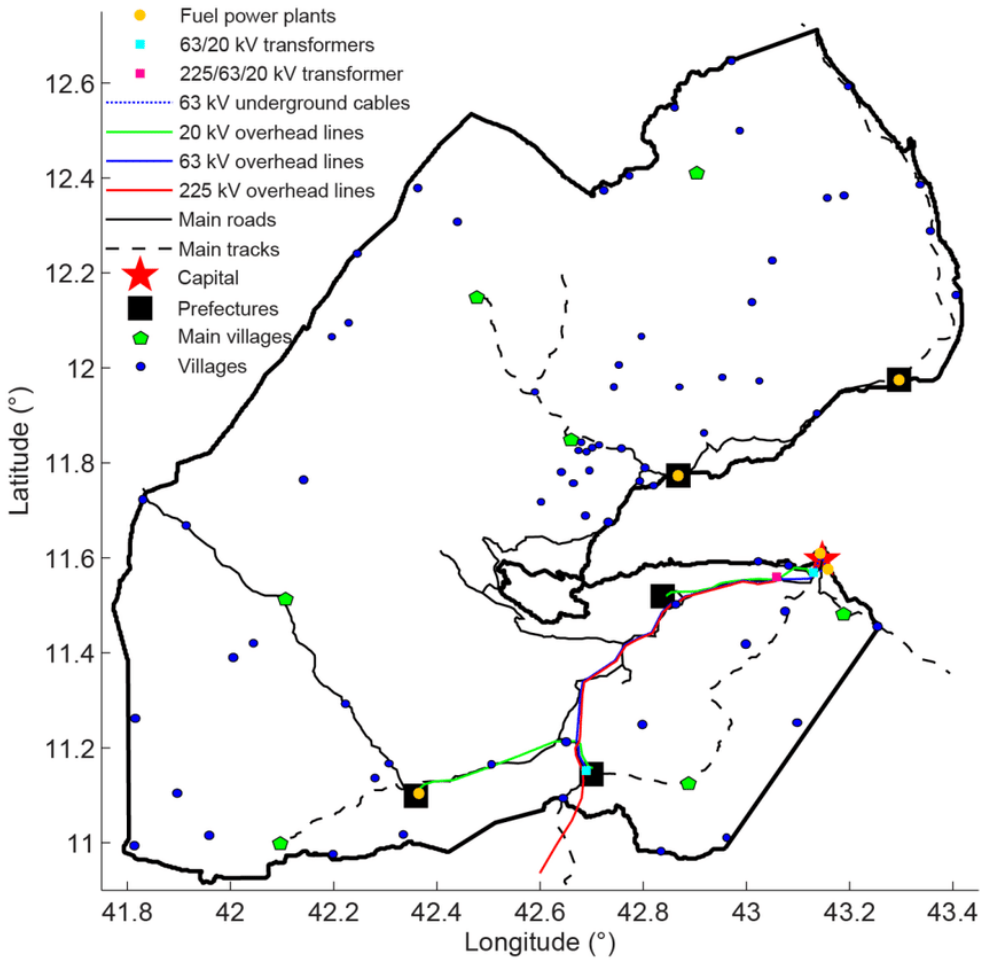
The grid emission factor ($EF_{grid,CM,y}$) is calculated ex-ante as per the "Tool to calculate the emission factor for an electricity-system" (Version 7.0). The emission factor is not monitored during the crediting period of the project activity.

This methodological tool determines the CO_2 emission factor for the displacement of electricity generated by power plants in an electricity system, by calculating the "combined margin" emission factor (CM) of the electricity system. The CM is the result of a weighted average of two emission factors pertaining to the electricity system: the "operating margin" (OM) and the "build margin" (BM). The operating margin is the emission factor that refers to the group of existing power plants whose current electricity generation would be affected by project activity. The build margin is the emission factor that refers to the group of prospective power plants whose construction and future operation would be affected by the project activity.

The tool indicates six steps for the calculation of the combined margin (CM) emission factor:

STEP 1. Identify the relevant electricity systems.

The following map shows the geographical boundary of the Djiboutian grid.



As of 2021 (latest data available before starting date), the DNA of the host country has not published a delineation of the project electricity system and connected electricity systems. Thus, the project participant defines the "*project electricity system*" and interconnected systems as per Option 2 based on EDD delineation (national utility responsible for electricity dispatching).

Therefore, the relevant electric power system is the national grid of Djibouti, including all power plants physically connected through transmission and distribution lines to the project activity. It is managed by EDD, which annually aggregated power production and import summaries have been obtained and relied on for the establishment of the grid emission factor calculation.

The project electricity system is also connected to the Ethiopian power grid[7], together consisting of the "*connected electricity system*" - thus electricity transfers from a connected electricity systems to the project electricity system are defined as electricity imports while electricity transfers from the project electricity system to connected electricity systems are defined as electricity exports.

STEP 2. Choose whether to include off-grid power plants in the project electricity system (optional).

[7] with transmission constraints as per https://www.afdb.org/en/documents/multinational-ethiopia-djibouti-power-system-interconnection-project-semera-galafi-230kv-transmission-line-project-p-z1-fa0-179/p-z-fa0-180-esia-executive-summary

The following option is selected to calculate the operating margin and build margin emission factor:

- Option II: Both grid power plants and off-grid power plants are included in the calculation.

  - Option IIb: the default CO_2 emission factor and the default value of the electricity generated by the off-grid power plants can be applied, since the following conditions for this option are met:

    (a) The project activity is located in (i) a Least Developed Country (LDC); and

    (b) The project activities consist of grid-connected renewable power generation; and

    (c) There is a load shedding program in place to compensate the deficit of the generation capacities[8].

For the off-grid power plants that choose Option IIb the default value of 0.8 tCO_2/MWh can be used for the CO_2 emission factor.

STEP 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on the following method, which is described under Step 4:

(b) Simple adjusted OM;

The Simple OM method (a) can be used when low-cost/must run resources constitute less than 50% of the total amount of the power generation on the grid, in average of the five most recent years. Between 2019 and 2023 the low-cost/must run share represented more than 50%. Therefore, method (a) Simple OM is inapplicable.

The simple adjusted OM method is applied since:

- LCMR share is greater than 50% in recent 5 years
- Average load by LCMR is greater than average LASL over three years
- Hourly loads of the grid in MW are unavailable and LASL > 1/3 HASL

Table 6: Net power generation and share of low cost/must run power plants

	5 years of historical data				
Year	**2019**	**2020**	**2021**	**2022**	**2023**
Net production [MWh]	494,585	522,713	532,304	616,684	573,894
Total share of low-cost/must-run [%]	79.0%	79.4%	78.4%	82.7%	79.4%

[8] Load-shedding embedded in IPP concessions ('délestage' in French as per http://www.droit-afrique.com/uploads/Djibouti-Decret-2019-13-activites-producteurs-independants-electricite.pdf as well as in EPC Art. 287. §1. 'Plan de délestage', load shedding program being an integral part of the 'Code de sauvegarde' (network connection contract)]

The simple adjusted OM emission factor ($EF_{grid,OM-adj,y}$) is thus applied. It is a variation of the simple OM, where the power plants/units (including imports) are separated in low-cost/must-run power sources (k) and other power sources (m).

STEP 4. Calculate the operating margin emission factor according to the selected method.

According to the tool, the simple adjusted OM emission factor ($EF_{grid,OM-adj,y}$) is calculated based on the net electricity generation of each power unit and an emission factor for each power unit, as follows:

$$EF_{grid,OM-adj,y} = (1-\lambda_y)\times\frac{\sum_m EG_{m,y}\times EF_{EL,m,y}}{\sum_m EG_{m,y}} + \lambda_y\times\frac{\sum_k EG_{k,y}\times EF_{EL,k,y}}{\sum_k EG_{k,y}}$$

Where:

$EF_{grid,OM-adj,y}$	=	Simple adjusted operating margin CO_2 emission factor in year y (tCO_2/MWh).
λ_y	=	Factor expressing the percentage of time when low-cost/must-run power units are on the margin in year y
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh).
$EG_{k,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit k in year y (MWh).
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
$EF_{EL,k,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units.
k	=	All low-cost/must run grid power units serving the grid in year y
y	=	The relevant year as per the data vintage chosen in Step 3.

Under this option, $EF_{EL,m,y}$, $EF_{EL,k,y}$, $EG_{m,y}$ and $EG_{k,y}$ are determined using the same procedures as those for the parameters $EF_{EL,m,y}$ and $EG_{m,y}$ in Option A of the simple OM method. For grid power plants, $EG_{m,y}$ is determined as per the provisions in the monitoring tables of the tool, i.e. for the ex-ante option "once for each crediting period using the most recent three historical years for which data [from utility or government records or official publications] is available at the time of submission of the PDD to the DOE for validation."

Since data on fuel consumption is not available, the emission factor ($EF_{EL,m,y}$) is determined as follows (option A2 of the tool):

$$EF_{EL,m,y} = \frac{EF_{co2,m,i,y}\times 3.6}{\eta_{m,y}}$$

Where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
$EF_{CO2,m,i,y}$	=	Average CO_2 emission factor of fuel type i used in power unit m in year y ($t\ CO_2/GJ$)

$\eta_{m,y}$		Average net energy conversion efficiency of power unit m in year y (ratio)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen in Step 3.
3,6	=	Conversion factor (GJ/MWh)

Off-grid power plants are included in the operating margin emission factor. The following default values can be used to determine $EG_{m,y}$ for the off-grid plants:

(a) The value of 10 per cent of the total electricity generation by grid power plants in the electricity system for the purpose of the operating margin determination[9];

CO_2 emission factor(s) for net electricity imports from the connected electricity system (Ethiopia) is conservatively considered as per TOOL07 §25 (a) as: 0 t CO_2/MWh. No imports from connected electricity systems located in Annex I country(ies) nor electricity exports occur in Djibouti.

Approach to determine lambda (λy) is chosen as Approach 1. Use default values of lambda from Table 1 appendix 2 based on the share of electricity generation from low-cost/must-run in total generation derived using 1) average of the five most recent years.

Approach 1 is applied because the LASL is not less than one-third of the HASL in a project electricity/ grid system demonstrated based on the yearly data for the years used to determine the OM emission factor.

Based on years 2021-2023, the calculated **Operating Margin is: 0.4205 tCO_2/MWh.**

STEP 5. Calculate the build margin (BM) emission factor.

No data to calculate the Build Margin is available because:

- There are no 5 most recent power units
- The Djibouti grid power units do not comprise at least 20 per cent of the system generation

Since the data requirements for the application of Step 5 above cannot be met, and the project activity is located in: (i) a Least Developed Country (LDC), Simplified CM = OM

STEP 6. Calculate the combined margin (CM) emission factor.

The combined margin emission factor is calculated following the simplified CM approach, because the grid is located in LDC/SIDs/URC.

$EF_{grid,CM,y} = EF_{grid,OM,y}$

Based on years 2021-2023, the **combined margin emission factor** and grid emission factor value used to calculate the emission reductions of the wind power plant project is **0.4205 tCO_2/MWh**

[9] Since scenario (b) relies on 'the sample group as per Step 5 for the purpose of the build margin determination', yet build margin is not applied (Simplified CM approach)

4.2 Project Emissions

Describe the procedure for quantification of project emissions and/or carbon stock changes in accordance with the applied methodology. Project emissions may be negative where carbon stock increases (sinks) exceed project emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

For most renewable energy power generation project activities, PE_y = 0. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y} + PE_{PSP,y}$$

Where:

PE_y	=	Project emissions in year *y* (t CO_2e/yr)
$PE_{FF,y}$	=	Project emissions from fossil fuel consumption in year *y* (t CO_2/yr)
$PE_{GP,y}$	=	Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO_2e/yr)
$PE_{HP,y}$	=	Project emissions from water reservoirs of hydro power plants and pumped storage plants in year y (t CO_2e/yr)
$PE_{BESS,y}$	=	Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (t CO_2e/yr)
$PE_{PSP,y}$	=	Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (t CO_2e/yr)

$PE_{FF,y}$, $PE_{GP,y}$, $PE_{HP,y}$, $PE_{BESS,y}$ and $PE_{PSP,y}$ are equal to 0 as the project is an installation of a wind power plant with no auxiliary fossil fuel consumption. In particular, ACM0002 §42 stipulates that for all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected.

4.3 Leakage Emissions

No other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, transport etc.) are neglected.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
01-Jan-2024 to 31-Dec-2024	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2025 to 31-Dec-2025	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2026 to 31-Dec-2026	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2027 to 31-Dec-2027	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2028 to 31-Dec-2028	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2029 to 31-Dec-2029	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2030 to 31-Dec-2030	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2031 to 31-Dec-2031	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2032 to 31-Dec-2032	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
01-Jan-2033 to 31-Dec-2033	*101,757*	*0*	*0*	*101,757*	*0*	*101,757*
Total	1,017,574	*0*	*0*	1,017,574	*0*	1,017,574

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” v7.0
Source of data	Electricité De Djibouti grid data (Données Energétique EDD 2022.xlsx)
Value applied	0.4205
Justification of choice of data or description of measurement methods and procedures applied	As per the “Tool to calculate the emission factor for an electricity system” v7.0
Purpose of data	• Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2/MWh
Description	Operating Margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” v7.0
Source of data	Electricité De Djibouti grid data (Données Energétiques EDD 25.09.24.xlsx and Grid EF 2023 simple-adjusted OM.xlsx
Value applied	0.4205
Justification of choice of data or description of measurement methods and procedures applied	As per the “Tool to calculate the emission factor for an electricity system” v7.0
Purpose of data	Calculation of baseline emissions

Comments	-

Data / Parameter	$EF_{CO2,i,y}$
Data unit	t CO_2/GJ
Description	CO2 emission factor of fuel type i used in power unit m in year y
Source of data	IPCC default values at the lower limit of uncertainty at a 95 per cent confidence interval as provided in table 1.4 of Chapter1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Value applied	75.5
Justification of choice of data or description of measurement methods and procedures applied	*Residual Fuel Oil default value*
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EG_{m,y}$, EG_{y}, $EG_{k,y}$
Data unit	MWh
Description	Net electricity generated by power plants/unit m, k in year y
Source of data	Utility or government records (Electricité De Djibouti)
Value applied	As per grid emission factor spreadsheet data
Justification of choice of data or description of measurement methods and procedures applied	Once for each crediting period using the most recent three historical years for which data is available at the time of submission of the CDM-PDD to the DOE for validation (ex-ante option);
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\eta_{m,y}$

Data unit	-
Description	Average net energy conversion efficiency of power unit m in year y
Source of data	The default values provided in in Table 2, Appendix of TOOL09:"Determining the baseline efficiency of thermal or electric energy generation systems"
Value applied	Share of LCMR: 81.86 per cent to 84.76 per cent — Lambda: 0.45 Share of LCMR: 78.72 per cent to 81.86 per cent — Lambda: 0.4
Justification of choice of data or description of measurement methods and procedures applied	Once for the crediting period
Purpose of Data	Calculation of baseline emissions
Comments	-

Value applied (nested table):

Share of LCMR	Lambda
81.86 per cent to 84.76 per cent	0.45
78.72 per cent to 81.86 per cent	0.4

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/yr
Description	Provide a brief description of the data/parameter
Source of data	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Description of measurement methods and procedures to be applied	Electricity output will be electronically stored by the Technical/Engineering/ Maintenance Department under the Plant Manager's authority
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	*242,000*
Monitoring equipment	*Bidirectional electricity meters*

QA/QC procedures to be applied	Calibration of the meters will be carried out as per manufacturer specifications. Bi-annual inspection and testing of accuracy will be carried out by the project proponent and EDD as per PPA. *Accuracy class: 0.2S for all meters (EDD and RSP)*
Purpose of data	Calculation of baseline emissions
Calculation method	Electronic recording
Comments	• *There are 4 meters in both substations (8 meters in total).* • *One main meter on each production line with its backup.* • *The meters on EDD side will be used for the invoicing (as agreed in the Grid Connection Agreement)* *The meters on RSP side are there to double-check* At the date of writing the PD, EDD has not installed its meters yet.

5.3 Monitoring Plan

The proposed project activity monitoring plan complies with the methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources (Version 20.0), whereby it is stated that:

"All data collected as part of monitoring should be archived electronically and be kept at least for 2 years after the end of the last crediting period. 100% of the data should be monitored if not indicated otherwise in the tables of Section 6.1 of ACM0002 Ver. 22. All measurements should be conducted with calibrated measurement equipment according to relevant industry standards".

Therefore, the quantity of net electricity generation supplied by the project plant to the grid is reliably monitored through calibrated electricity meters and cross-checked with sales records as per internal monitoring practices.

As concerns the monitoring, Red Sea Power management has the overall responsibility for all VCS monitoring of the project, including:

- Develop, approve, execute, and improve the VCS Monitoring/Reporting Procedures;
- Organize in-house seminar to inform and train the company staff to the monitoring procedures;
- Ensure that instrumentations and devices are available and properly suited to efficiently perform the monitoring;

- Communicate and coordinate the monitoring work of all business units;
- Validate and electronically archive all monitoring data on a monthly basis throughout the crediting period (and conserve it at least for 2 further years);
- Calculate and report the emission reductions; and
- Coordinate the DOE work during the verification audit.

A VCS coordinator might be appointed with delegation of the above specific tasks of monitoring supervision. Recorded data is immediately collected and managed in user-friendly, detailed reports and tables to facilitate analysis.

Monitoring team and training

Data collection, consolidation and results analysis are undertaken by a dedicated team adequately trained, well aware of VCS requirements. This team does not have any hierarchical relationships or dependence links with all entities involved to measure net electricity supplied to the grid and to assure the correct operation and maintenance of the measuring equipment. This independence shall guarantee the integrity of the work that is done.

Emergency and trouble-shooting procedures

Immediate response capability is achieved through the implementation of a system of supervision and control that transfers real-time all information about the state of the equipment. The operator will provide a team of qualified stakeholders that can react in "real time" according to unscheduled outage and maintenance procedures.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.